Name

Mathematics Extension 1 Year 11

Parametric Form

Book 1

Version: 261028

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Contents

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Learning Intention ..... **Error! Bookmark not defined.**

Syllabus Content

ME1-11-01 solves problems involving inequalities, functions and their inverses, graphical relationships between functions, and parametric equations

Parametric form of a function or relation

- Recognise that a curve may be represented by two parametric equations that give x and y as functions of a parameter and that the curve may also have a Cartesian equation in x and y
- Express linear and quadratic functions and circles in parametric form
- Convert linear and quadratic functions and circles from parametric form to Cartesian form
- Graph linear and quadratic functions and circles expressed in parametric form

Parametric Form

ⓘ Parametric Form of an Equation

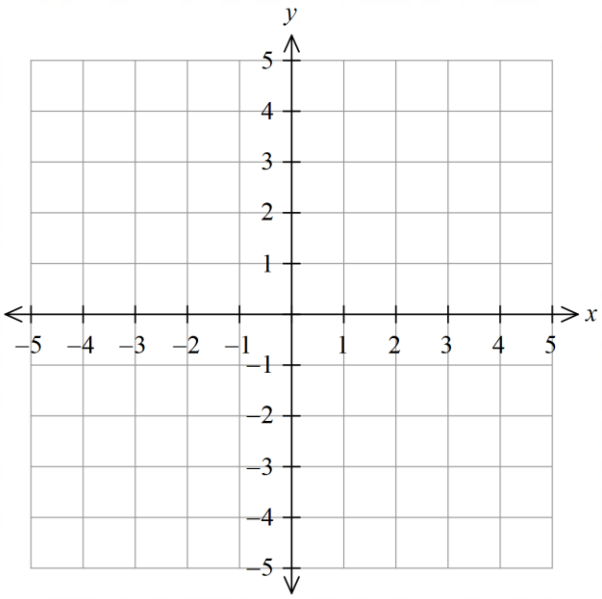
- **Cartesian equations** directly link x and y in one equation (like $y = 2x + 1$).
- In **parametric equations**, x and y are defined separately using a third variable (the **parameter**). Instead of y depending on x, both x and y depend on the parameter.
- Parametric equations are useful as they track multiple changing values at once. They have essential applications in future calculus topics such as projectile motion.

🔍 Sketching a curve from its parametric equations

a $x = t + 2$
$y = 2t$

t	-2	-1	0	1	2
x					
y					

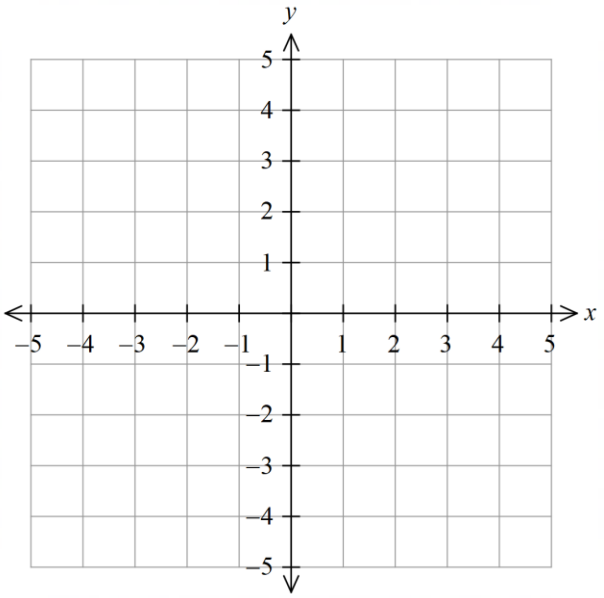
Express these parametric equations as a single Cartesian equation.



b $x = 2t$
$y = 2t^2$

t	-2	-1	0	1	2
x					
y					

Express these parametric equations as a single Cartesian equation.



ⓘ Converting from Parametric Form to Cartesian Form

- To convert parametric equations to Cartesian form, we need to eliminate the parameter:
  1. Rearrange one of the equations so that the parameter is the subject.
  2. Substitute the parameter into the other equation.

Example Convert from parametric form to Cartesian form

Worked example	Your Turn
Find the Cartesian equation. $x = 3t + 1$ (1) $y = 2t - 3$ (2) 1. Rearrange Using (1): $t = \frac{x-1}{3}$ 2. Substitute Sub $t = \frac{x-1}{3}$ into (2) $y = 2\left(\frac{x-1}{3}\right) - 3$ $3y = 2(x-1) - 9$ $3y = 2x - 2 - 9$ $3y = 2x - 11$ $2x - 3y - 11 = 0$	Find the Cartesian equation. $x = 2t - 1$ $y = 3t + 2$ $3x - 2y + 7 = 0$
Find the Cartesian equation. $x = 2p$ (1) $y = p^2 - 3$ (2) 1. Rearrange Using (1): $p = \frac{x}{2}$ 2. Substitute Sub $p = \frac{x}{2}$ into (2) $y = \left(\frac{x}{2}\right)^2 - 3$ $y = \frac{x^2}{4} - 3$ $y = \frac{x^2}{4} - 3$	Find the Cartesian equation. $x = 4p$ $y = p^2 + 1$ $y = \frac{x^2}{16} + 1$

ⓘ Converting from Parametric Form to Cartesian Form

- To convert Cartesian equations to parametric form, we need to define a parameter. How to define the parameter will be given by the question.
  1. Substitute the parametric equation of x into the Cartesian equation.
  2. Rearrange to make y the subject and simplify.
  3.

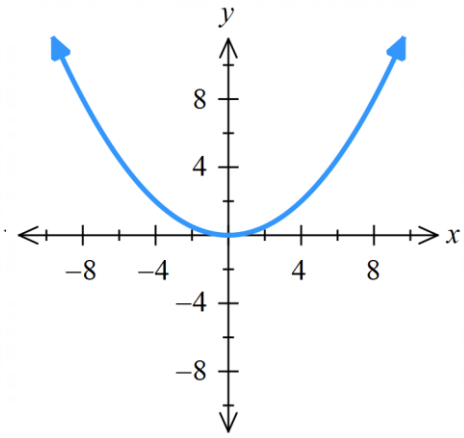
Example Convert from Cartesian form to parametric form

Worked example	Your Turn
$5x - 2y + 10 = 0$, using $x = 4t$ $5(4t) - 2y + 10 = 0$ $20t - 2y + 10 = 0$ $2y = 20t + 10$ $y = 10t + 5$ $\therefore x = 4t,$ $y = 10t + 5$	$3x + y - 3 = 0$, using $x = 1 - t$ $x = 1 - t,$ $y = 3t$
$y = x^2 + 3x - 1,$ using $x = 2t - 2$ $y = (2t - 2)^2 + 3(2t - 2) - 1$ $y = 4t^2 - 8t + 4 + 6t - 6 - 1$ $y = 4t^2 - 2t - 3$ $\therefore x = 2t - 2,$ $y = 4t^2 - 2t - 3$	$x^2 = 4(y - 3)$, using $x = 2t$ $x = 2t,$ $y = t^2 + 3$

1

a Complete the table for $x = 4t$, $y = 2t^2$.

t	-2	-1	0	1	2
$x = 4t$	-8	-4	0	4	8
$y = 2t^2$	8	2	0	2	8



b Sketch the graph.

c Eliminate the parameter to find the Cartesian equation.

$y = \frac{x^2}{8}$

d The curve is a parabola. What value of t gives the coordinates of the vertex?

$t = 0$

2 Consider the curve defined parametrically by $x = 3t$, $y = t^2$.

a Complete the table.

t	-3	-2	-1	0	1	2	3
$x = 3t$	-9	-6	-3	0	3	6	9
$y = t^2$	9	4	1	0	1	4	9

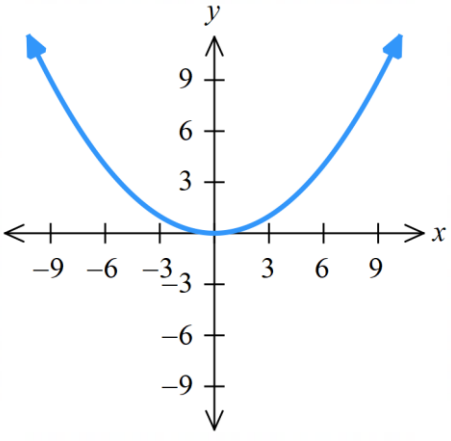
b Eliminate the parameter and find the Cartesian equation.

$y = \frac{x^2}{9}$

c What value of t would give the coordinate $(6,4)$?

$t = 2$

d Sketch the curve.



3

a Show that the point $\left(cp, \frac{c}{p}\right)$ lies on the hyperbola $xy = c^2$, where c is a constant.

$xy = (cp) \cdot \left(\frac{c}{p}\right)$

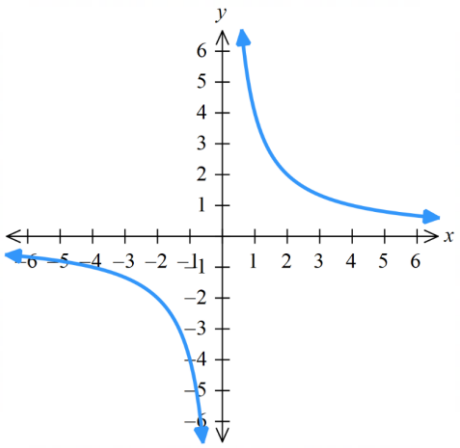
$= cp \cdot \frac{c}{p}$

$= c^2$

b Complete the table of values for $x = 2p, y = \frac{2}{p}$.

p	-3	-2	-1	$-\frac{1}{2}$	0	$\frac{1}{2}$	1	2	3
$x = 2p$	-6	-4	-2	-1	0	1	2	4	6
$y = \frac{2}{p}$	$-\frac{2}{3}$	-1	-2	-4	Undef.	4	2	1	$\frac{2}{3}$

c Sketch the graph.



4 Consider the parametric equations $y = 2t - 1$ and $x = t - 2$.

a Complete the table.

t	-2	-1	0	1	2
$x = t - 2$	-4	-3	-2	-1	0
$y = 2t - 1$	-5	-3	-1	-1	3

b Explain, using the table, why the graph is a straight line.

As x changes, y changes by a constant amount.

In this case, as x increases by 1, y increases by 2.

Therefore, this is a straight line.

c From the table, find the y-intercept and gradient.

Using any two points, gradient is 2.

Using last column, when $x = 0, y = 3$ so $y-$intercept is 3.

d Eliminate t to find the Cartesian equation and check it with part **c**.

$t = x + 2$

$y = 2(x - 2) - 1$

$y = 2x + 3$

5 Eliminate the parameter and find the Cartesian equation of the curve.

Hint: use $a^2 + b^2 = (a+b)^2 - 2ab$

$$x = p + \frac{1}{p}$$

$$y = p^2 + \frac{1}{p^2}$$

Using identity $a^2 + b^2 = (a+b)^2 - 2ab$:

$$y = \left(p + \frac{1}{p}\right)^2 - 2$$

$$\therefore y = x^2 - 2$$

6 Write in Cartesian form.

a $x = 2t,$ $y = 3t - 4$

$3x - 2y - 8 = 0$

b $x = 3n - 2,$ $y = 2n - 5$

$2x - 3y - 11 = 0$

c $x = -6t,$ $y = 3t + 2$

$x + 2y - 4 = 0$

d $x = r - 3$ $y = r^2 + 1$

$y = x^2 - 3x + 6$

e $x = 2p + 3,$ $y = p^2$

$x^2 - 6x - 4y + 9 = 0$

f $x = 4k - 7,$ $y = 2k^2 + k$

$x^2 + 16x - 8y + 63 = 0$

7 Write each function in parametric form, using the given equation for x.

a $3x - 4y + 24 = 0,$ $x = 4t$

$x = 4t, y = 3t + 6$

b $y = x^2,$ $x = 2t - 1$

$x = 2t - 1, y = 4t^2 - 4t + 1$

c $y = 2x^2 - 3x + 4,$ $x = 1 - t$

$x = 1 - t, y = 2t^2 - t + 3$

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The parametric equations of a line are given by $x = 1 + 3t,\ y = 4t$.

Find the gradient and y-intercept of this line.

$t = \frac{x-1}{3}$

$y = 4\left(\frac{x-1}{3}\right)$

$y = \frac{4}{3}x - \frac{4}{3}$

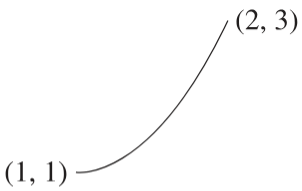
Gradient $\frac{4}{3}$, y-int $\frac{4}{3}$

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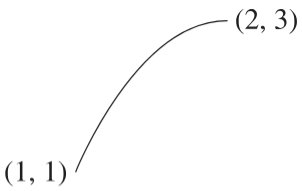
A curve is defined by $x = 2 + t$ and $y = 3 - 2t^2$ for $-1 \leq t \leq 0$.

Which of these best represents the curve?

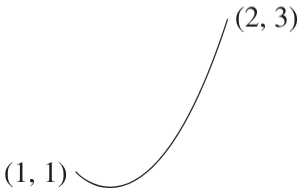
A.



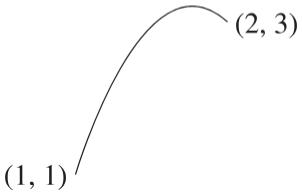
B.



C.



D.



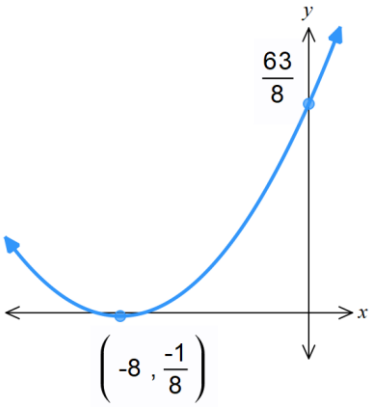
Cartesian equation is $y = 3 - 2(x-2)^2$

Concave down parabola with maximum at (2,3)

$\therefore B$

10 HSC Extension 1 Sample Question Band 4

a Sketch the graph of the function defined by $x = 4t - 7, y = 2t^2 + t$.



$t = \frac{x+7}{4}$

$y = 2\left(\frac{x+7}{4}\right)^2 + \left(\frac{x+7}{4}\right)$

This simplifies to

$8y = x^2 + 16x + 63$

$y = \frac{1}{8}(x+7)(x+9)$

Turning point at $x = -8, y = -\frac{1}{8}$

b State the domain and range of the function.

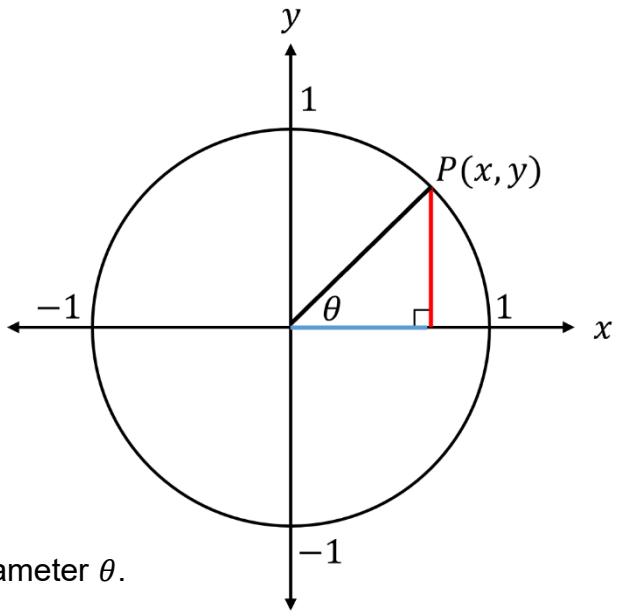
$x \in \mathbb{R}, y \geq -\frac{1}{8}$

Parametric Equation of a Circle

Parametric Equation of a Circle

From your knowledge of trigonometry,
remember that we can define a unit circle with:

$x = \cos\theta$
$y = \sin\theta$



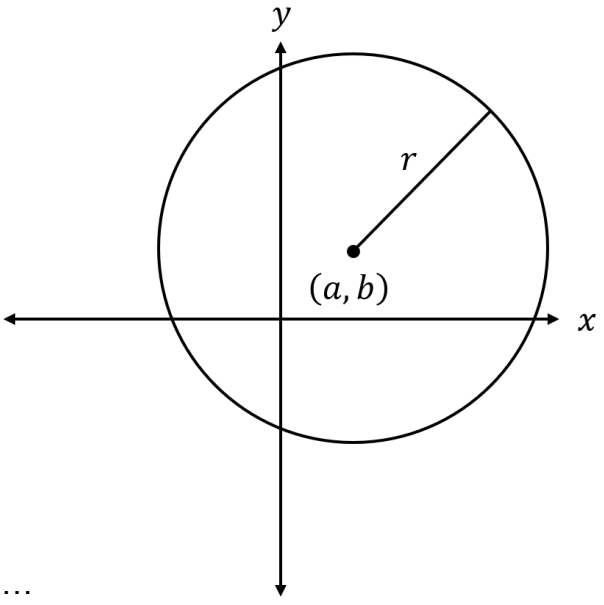
we can use Pythagoras' Theorem to write:

$$x^2 \quad + \quad y^2 \quad = 1$$

$$\ldots\ldots\ldots + \ldots\ldots\ldots = 1$$

It shows that we can define a circle using the parameter θ.
Note that this is the Pythagorean identity.

Consider a circle with centre (a, b) and radius r.



$$(x-a)^2 + (y-b)^2 = r^2$$

Divide by r^2

$$\ldots\ldots\ldots + \ldots\ldots\ldots = 1$$

By relating this to $\cos^2\theta + \sin^2\theta = 1$.

We should define:

$\cos\theta = \ldots\ldots\ldots\ldots\ldots$ and $\sin\theta = \ldots\ldots\ldots\ldots\ldots$

$\therefore x = \ldots\ldots\ldots\ldots\ldots$ and $y = \ldots\ldots\ldots\ldots\ldots$

ⓘ Parametric Form of a Circle

Circle centred at the origin	Circle centred at (a, b)
Cartesian equation: $$x^2 + y^2 = r^2$$ Parametric equations: $$x = r\cos\theta$$ $$y = r\sin\theta$$	Cartesian equation: $$(x-a)^2 + (y-b)^2 = r^2$$ Parametric equations: $$x = r\cos\theta + a$$ $$y = r\sin\theta + b$$

- To convert the parametric form of a circle to Cartesian form:
  1. Rearrange given equations to make $\cos\theta$ and $\sin\theta$ the subject.
  2. Substitute into $\cos^2\theta + \sin^2\theta = 1$.

Example Convert parametric form of a circle to Cartesian form

Worked Example

Find the Cartesian equation of a curve whose parametric equations are:

$x = 2\cos\theta \qquad y = 2\sin\theta$

1. Rearrange

$\cos\theta = \frac{x}{2} \qquad \sin\theta = \frac{y}{2}$

2. Substitute into $\cos^2\theta + \sin^2\theta = 1$

$$\left(\frac{x}{2}\right)^2 + \left(\frac{y}{2}\right)^2 = 1$$
$$\frac{x^2}{4} + \frac{y^2}{4} = 1$$
$$x^2 + y^2 = 4$$

Your Turn

Find the Cartesian equation of a curve whose parametric equations are:

$x = 5\cos\theta \qquad y = 5\sin\theta$

$x^2 + y^2 = 25$

Find the Cartesian equation of a curve whose parametric equations are:

$x = 4 + 3\cos\theta \qquad y = 3 + 3\sin\theta$

1. Rearrange

$\cos\theta = \frac{x-4}{3} \qquad \sin\theta = \frac{y-3}{3}$

2. Substitute $\cos^2\theta + \sin^2\theta = 1$

$$\left(\frac{x-4}{3}\right)^2 + \left(\frac{y-3}{3}\right)^2 = 1$$
$$\frac{(x-4)^2}{9} + \frac{(y-3)^2}{9} = 1$$
$$(x-4)^2 + (y-3)^2 = 9$$

Find the Cartesian equation of a curve whose parametric equations are:

$x = 3 + 2\cos\theta \qquad y = 2\sin\theta - 4$

$(x-3)^2 + (y+4)^2 = 4$

ⓘ Parametric Form of a Circle

- To convert the Cartesian form of a circle to parametric form:
  1. Rewrite in the form $\left(\frac{x-a}{r}\right)^2 + \left(\frac{y-b}{r}\right)^2 = 1$
  2. Define $\cos\theta = \frac{x-a}{r}$ and $\sin\theta = \frac{y-b}{r}$
  3. Rearrange to make x and y the subjects.

  You could also memorise $x = r\cos\theta + a$, $y = r\sin\theta + b$.

Example Convert Cartesian form of a circle to parametric form

Worked Example	Your Turn
Write in parametric form. $(x-1)^2 + (y+2)^2 = 9$ 1. Rewrite = 1 $\frac{(x-1)^2}{9} + \frac{(y+2)^2}{9} = 1$ $\left(\frac{x-1}{3}\right)^2 + \left(\frac{y+2}{3}\right)^2 = 1$ 2. Define $\cos\theta = \frac{x-1}{3}$ $\sin\theta = \frac{y+2}{3}$ 3. Rearrange $x = 3\cos\theta + 1$ $y = 3\sin\theta - 2$	Write in parametric form. $(x+3)^2 + (y-2)^2 = 16$ $x = 4\cos\theta - 3$ $y = 4\sin\theta + 2$

1 Write in Cartesian form.

a $x = 6\cos\theta$, $y = 6\sin\theta$

$x^2 + y^2 = 36$

b $x = \sin\theta$, $y = \cos\theta + 1$

$x^2 + (y-1)^2 = 1$

c $x = 5 + 4\cos\theta$, $y = 2 + 4\sin\theta$

$(x-5)^2 + (y-2)^2 = 16$

2 Write each equation of a circle in parametric form with the parameter θ.

a $x^2 + y^2 = 4$

$x = 2\cos\theta$
$y = 2\sin\theta$

b $(x-3)^2 + (y+1)^2 = 36$

$x = 3 + 6\cos\theta$
$y = -1 + 6\sin\theta$

c $x^2 + 4x + y^2 - 5 = 0$

$x = -2 + 3\cos\theta$
$y = 3\sin\theta$

3

a Show that $x = a + r\cos\theta$ and $y = b + r\sin\theta$ define a circle with centre (a, b) and radius r.

Rearranging:

$\cos\theta = \frac{x-a}{r}$

$\sin\theta = \frac{y-b}{r}$

Using Pythagorean identity $\cos^2\theta + \sin^2\theta = 1$:

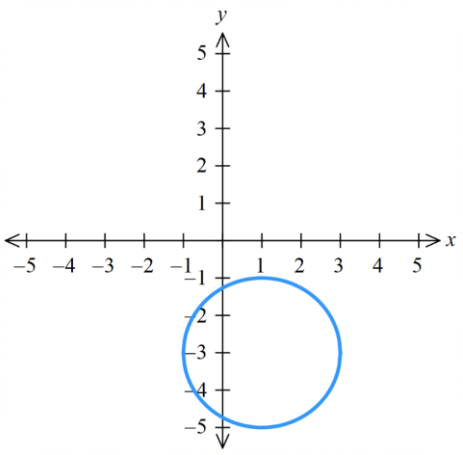
$\left(\frac{x-a}{r}\right)^2 + \left(\frac{y-b}{r}\right)^2 = 1$

Multiplying by r^2:

$(x-a)^2 + (y-b)^2 = r^2$

This is the equation of a circle with centre (a,b) and radius r.

b Hence sketch the curve $x = 1 + 2\cos\theta$, $y = -3 + 2\sin\theta$.
Show the intercepts.



$(x-1)^2 + (y+3)^2 = 4$, y-intercepts at $y = -3 \pm \sqrt{3}$

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Determine the Cartesian equation of the circle given by the parametric equations

$x = -3 + 4\cos\theta \quad y = 1 - 4\sin\theta$

A. $(x+3)^2 + (y-1)^2 = 4$

B. $(x-3)^2 + (y+1)^2 = 4$

C. $(x+3)^2 + (y-1)^2 = 16$

D. $(x-3)^2 + (y+1)^2 = 16$

C

5 HSC Sample Question Band 3

A circle has equation $x^2 - 10x + y^2 + 6y + 25 = 0$.

Express the circle in parametric form.

By completing the square, $(x-5)^2 + (y+3)^2 = 9$

$x = 5 + 3\cos\theta$

$y = -3 + 3\sin\theta$

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The curve given by $x = 3\sec t + 1$ and $y = 2\tan t - 1$ can be expressed in Cartesian form as:

A. $\dfrac{(y+1)^2}{4} - \dfrac{(x-1)^2}{9} = 1$

B. $\dfrac{(x+1)^2}{3} - \dfrac{(y-1)^2}{2} = 1$

C. $\dfrac{(x-1)^2}{3} + \dfrac{(y+1)^2}{2} = 1$

D. $\dfrac{(x-1)^2}{9} - \dfrac{(y+1)^2}{4} = 1$

$\sec t = \dfrac{x-1}{3}$

$\tan t = \dfrac{y+1}{2}$

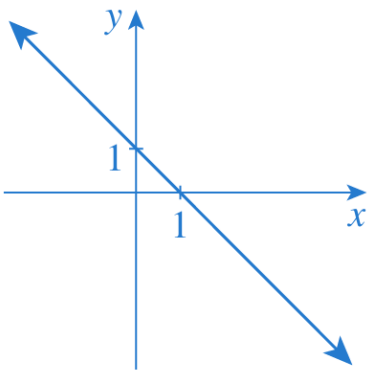
$\left(\dfrac{x-1}{3}\right)^2 - \left(\dfrac{y+1}{2}\right)^2 = 1$

$\dfrac{(x-1)^2}{9} - \dfrac{(y+1)^2}{4} = 1$

D

7 Different parametric representations may result in the same Cartesian equation.
The graphical representation may be different, due to domain and range restrictions.

a Find the Cartesian equation of $x = 2 - t$ and $y = t - 1$ and sketch the graph.



$t = 2 - x$

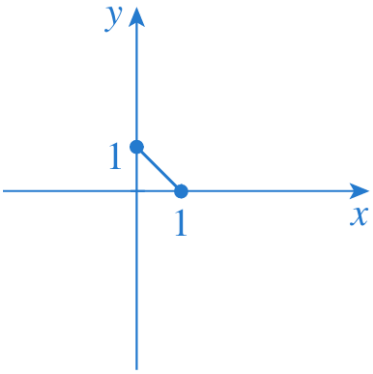
Substituting into $y = t - 1$:

$y = (2 - x) - 1$

$y = 1 - x$

$\therefore x + y = 1$

b Find the Cartesian equation of the curve $(\sin^2 t, \cos^2 t)$.
Explain why $0 \leq x \leq 1$ and $0 \leq y \leq 1$ and sketch the graph.



Let $x = \sin^2 t$ and $y = \cos^2 t$

Using the Pythagorean identity $\sin^2 t + \cos^2 t = 1$:

$x + y = 1$

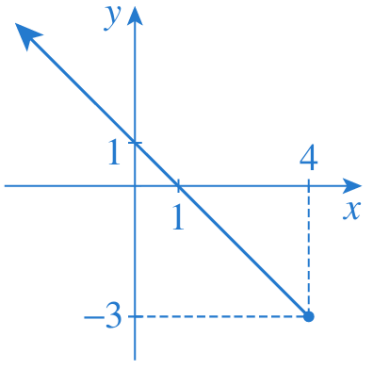
Domain and range restrictions:

Since $0 \leqslant \sin^2 t \leqslant 1$, $\therefore 0 \leqslant x \leqslant 1$

Since $0 \leqslant \cos^2 t \leqslant 1$, $\therefore 0 \leqslant y \leqslant 1$

The Cartesian equation is $x + y = 1$ with domain $0 \leqslant x \leqslant 1$ and $0 \leqslant y \leqslant 1$.

c Find the Cartesian equation of $x = 4 - t^2, y = t^2 - 3$.
Explain why $x \leq 4$ and $y \geq -3$ and sketch the curve.



From $x = 4 - t^2$, $t^2 = 4 - x$

Substituting into $y = t^2 - 3$:

$y = (4 - x) - 3$

$y = 1 - x$

$\therefore x + y = 1$

Domain and range restrictions:

Since $t^2 \geqslant 0$:

$x = 4 - t$; $x = 4 - t^2 \leq 4$

$y = t^2 - 3$; $y = t^2 - 3 \geq -3$

The Cartesian equation is $x + y = 1$ with $x \leqslant 4$ and $y \geqslant -3$.

8 Consider $(t^2, t^2 - 1)$. Jimmy claims that the graph is just the graph of $y = x - 1$,
whereas Jenny claims that it is only the right-hand side of the graph.
Determine who is correct, you can start by substituting some values of t and plotting the graph.

Mary is correct as she considers that $t^2 \geq 0$, so the domain is $x = t^2 \geq 0$.

9 Write in Cartesian form.

a $x = 1 + 2\tan\theta$
$y = 3\sec\theta - 4$

$\tan\theta = \frac{x-1}{2}$

$\sec\theta = \frac{y+4}{3}$

Using $\sec^2\theta - \tan^2\theta = 1$:

$\left(\frac{y+4}{3}\right)^2 - \left(\frac{x-1}{2}\right)^2 = 1$

$4(y+4)^2 - 9(x-1)^2 = 36$

b $x = \cos\theta + \sin\theta$
$y = \cos\theta - \sin\theta$

$x + y = 2\cos\theta \rightarrow \cos\theta = \frac{x+y}{2}$

$x - y = 2\sin\theta \rightarrow \sin\theta = \frac{x-y}{2}$

Using $\cos^2\theta + \sin^2\theta = 1$:

$\left(\frac{x+y}{2}\right)^2 + \left(\frac{x-y}{2}\right)^2 = 1$

$(x+y)^2 + (x-y)^2 = 4$

$2x^2 + 2y^2 = 4$

$\therefore x^2 + y^2 = 2$

$x^2 + y^2 = 2$

10 **2021 VCE Specialist Mathematics**

A function is defined parametrically by

$x(t) = 5\cos(2t) + 1, \quad y(t) = 5\sin(2t) - 1$

If $A(6,-1)$ and $B(1,4)$ are two points that lie on the function, then find the shortest distance along the graph from A to B.

$\cos 2t = \frac{x-1}{t}, \sin 2t = \frac{y+1}{5}$

$\cos^2(2t) + \sin^2(2t) = 1$

$(x-1)^2 + (y+1)^2 = 25$

A and B are the rightmost and topmost points on the circle, so the shortest arc connecting them is a quarter of a circle.

$d = \frac{2\pi r}{4} = \frac{5\pi}{2}$ units

11 **HSC Sample Question Band 5**

A curve has equation $\frac{(y-2)^2}{9} - \frac{(x+1)^2}{4} = 1$

Which of these is the curve in parametric form?

A. $x = 2\sec\theta - 1,\ y = 3\tan\theta + 2$

B. $x = 2\sin\theta - 1,\ y = 3\cos\theta + 2$

C. $x = 2\tan\theta - 1,\ y = 3\sec\theta + 2$

D. $x = 4\sec\theta - 1,\ y = 3\tan\theta + 2$

Try each option by substituting x and y, simplifying LHS and seeing which gives you 1.

Consider option C:

$\text{LHS} = \frac{(3\sec\theta + 2 - 2)^2}{9} - \frac{(2\tan\theta - 1 + 1)^2}{4}$

$= \frac{9\sec^2\theta}{9} - \frac{4\tan^2\theta}{4}$

$= \sec^2\theta - \tan^2\theta$

$= 1$

So C is the answer.